Making sense of soil - detecting soil change with multivariate sensing and mechanistically informed machine learning











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Soils

Sensors and data networking

Data science







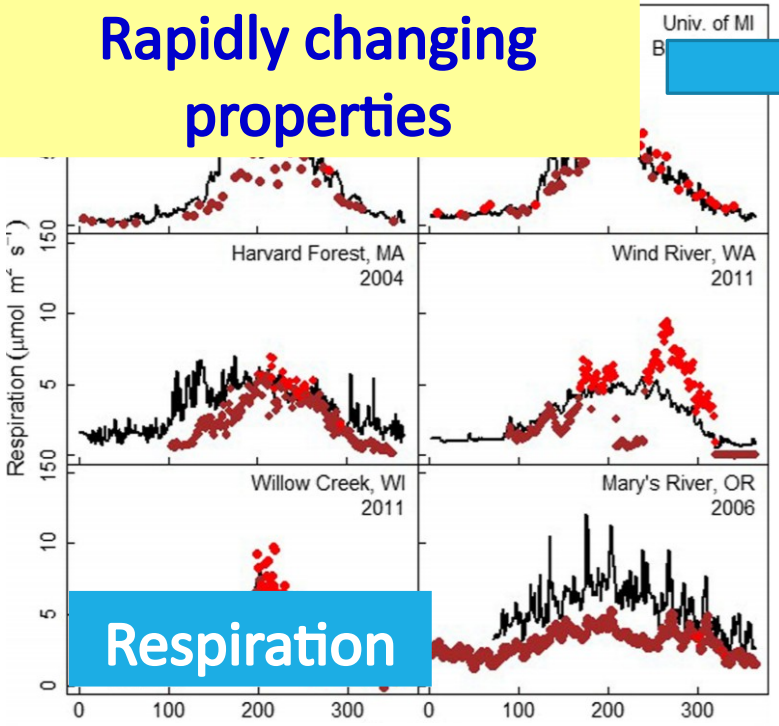




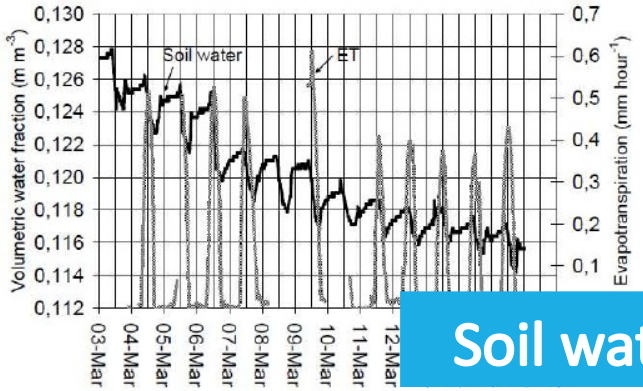
Hypothesis

Information contained in the high frequency signals from in-situ multi-measure and sensing configurations have sufficient information to predict the current state and direction of lower frequency soil properties that are critical to soil health.

Rapidly changing properties

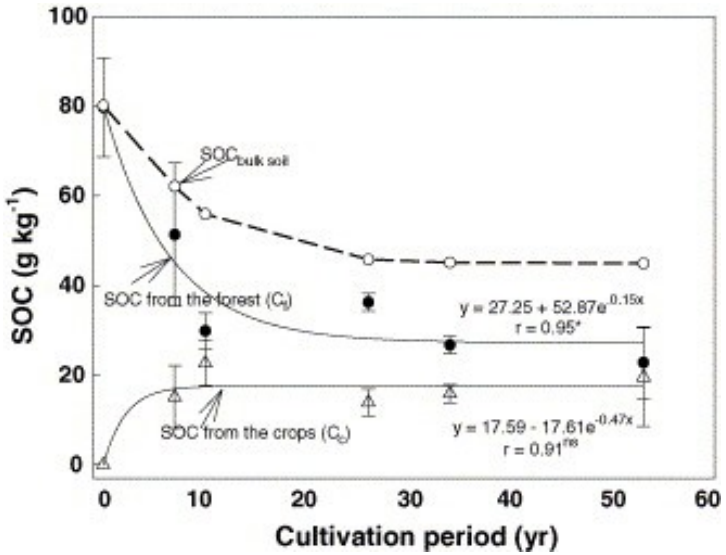


Respiration



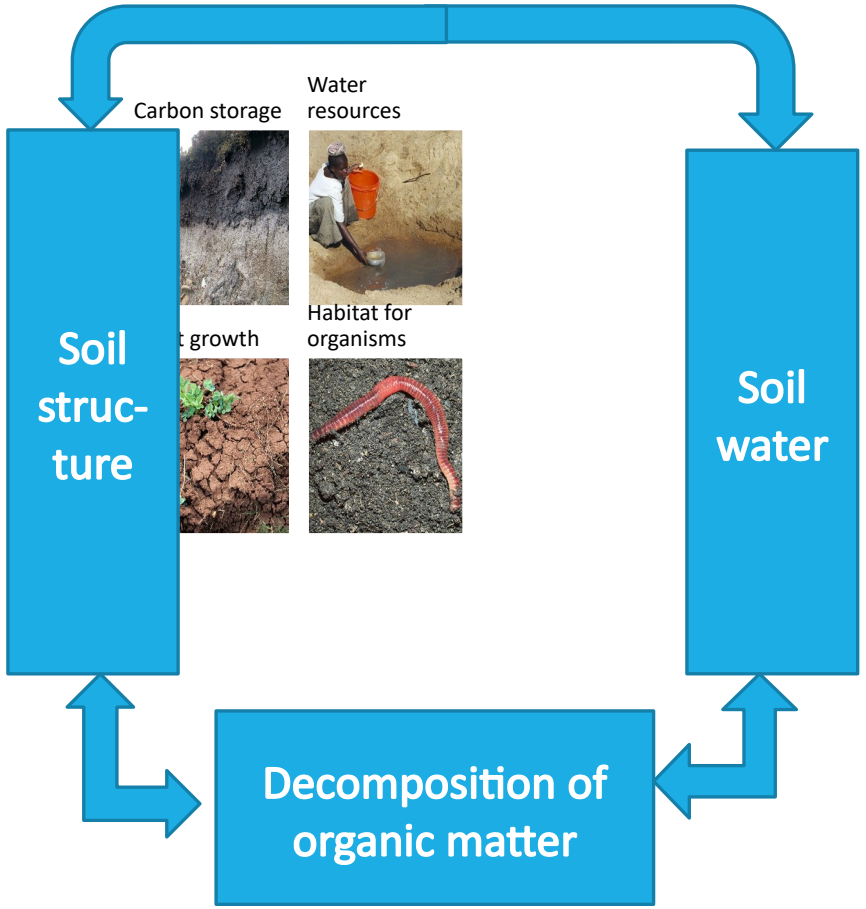
Soil water

Slowly changing properties



Soil carbon following forest clearance

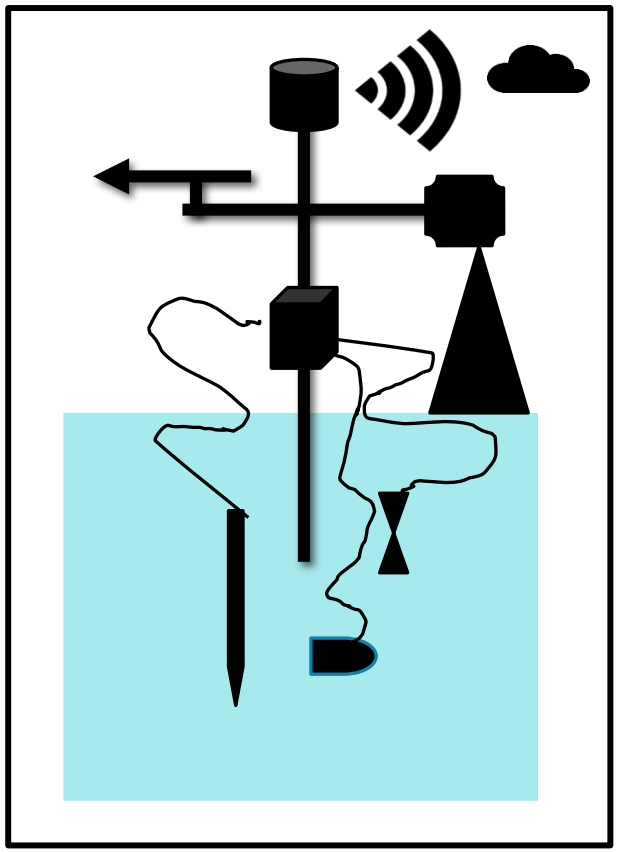
Soil multi-functionality



Leminih et al (2006) *Agriculture, Ecosystems & Environment* **109**, 9-19; Phillips et al. (2017)Plant and Soil **413**: 1-25; Archer et al (2012) Ecohydrology **5**: 108-120

Advance our understanding of dynamic soil processes that operate at different temporal/spatial scales.

1. Develop novel multi-functional soil sensing platforms
2. Integrate data streams and provide multi-functional prediction of rate and direction of soil restoration and degradation
3. Scale-up monitoring and predictive capacity



1. Multi-functional soil sensing platform



CONVENTIONAL SENSORS



LAB-PROVEN

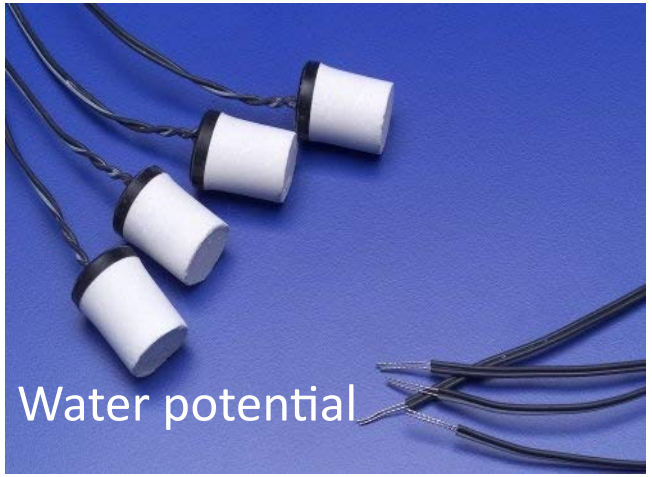


NOVEL DECOMPOSITION SENSORS

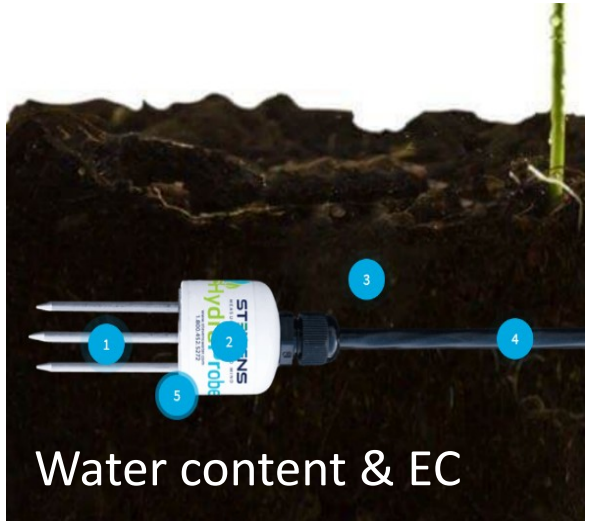


STANDARD SAMPLE COLLECTION

Conventional high frequency sensors up and running quickly

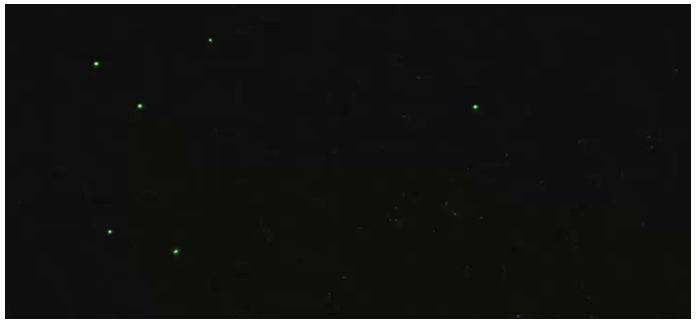


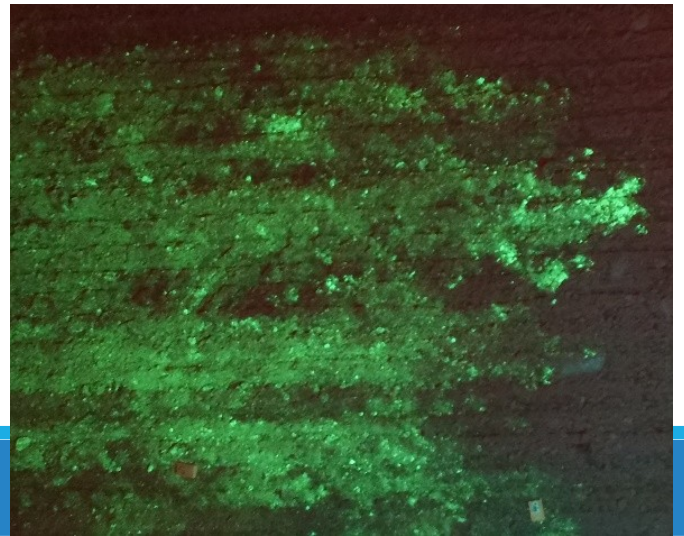




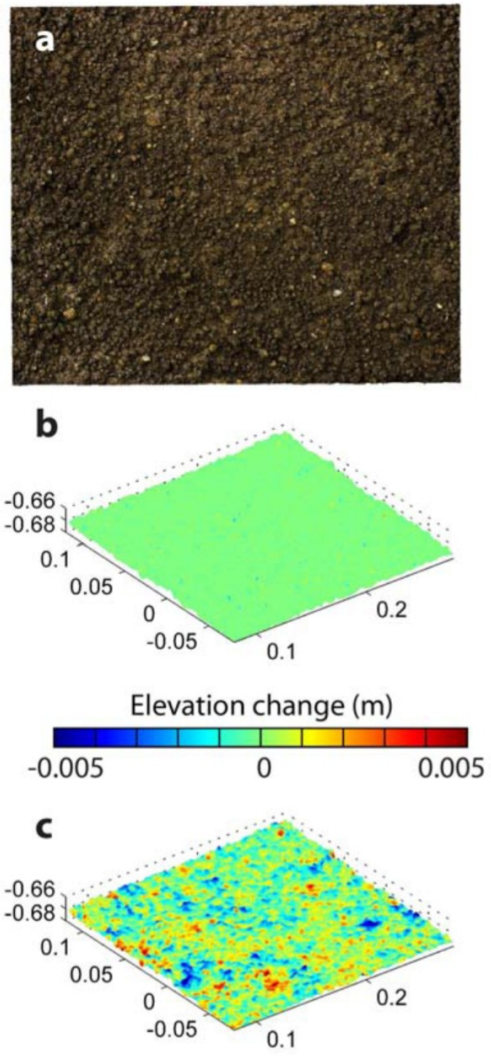
Lab proven high frequency sensors requiring more development work

Particle movement

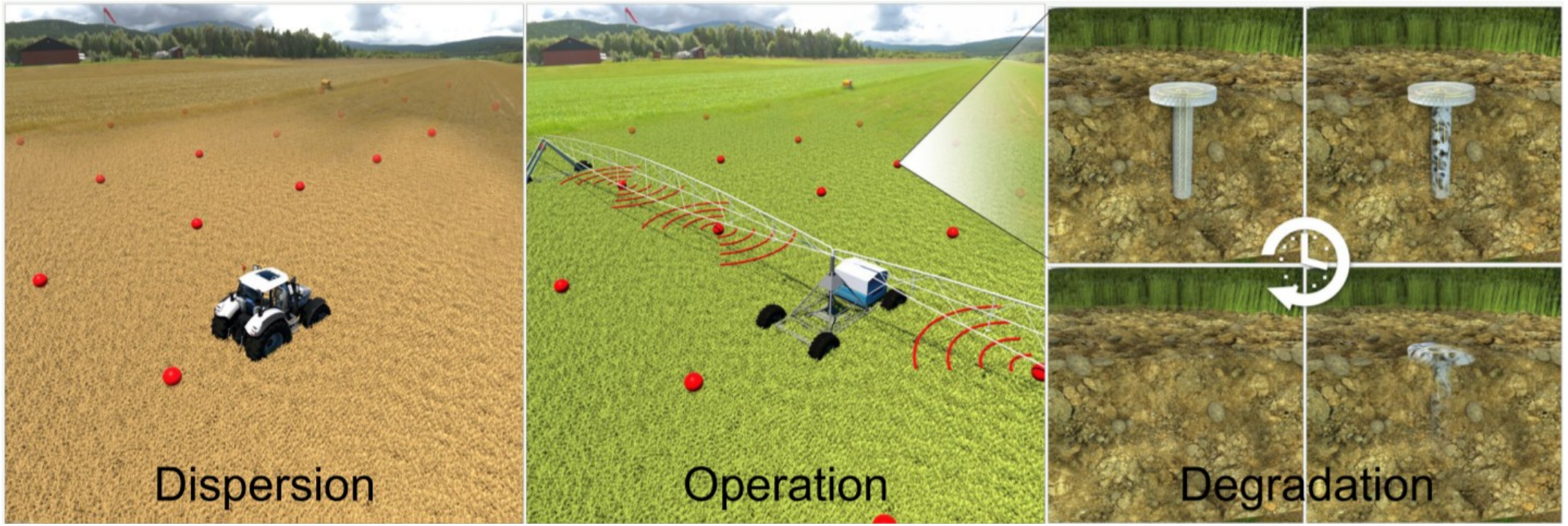




Structural change



Novel decomposition sensors



Precision Agriculture using Networks of Degradable Analytical Sensors (PANDAS)

low-cost additively manufactured, biodegradable sensors in order to provide high-spatial resolution mapping of various chemical analytes of interest in soil

Periodic sampling

- 20 soil functions
- Water regulation
- Erosion resistance
- Nutrient cycling
- Soil Carbon stocks
- Plant growth and quality



Boulder municipality rangeland restoration
Starts 2020

Colt Park meadow restoration experiment
Established 1989 Yorkshire Dales northern England (soils 1994):

2. Integrate data streams and provide multi-functional prediction of rate and direction of soil restoration and degradation

Process based vs machine learning

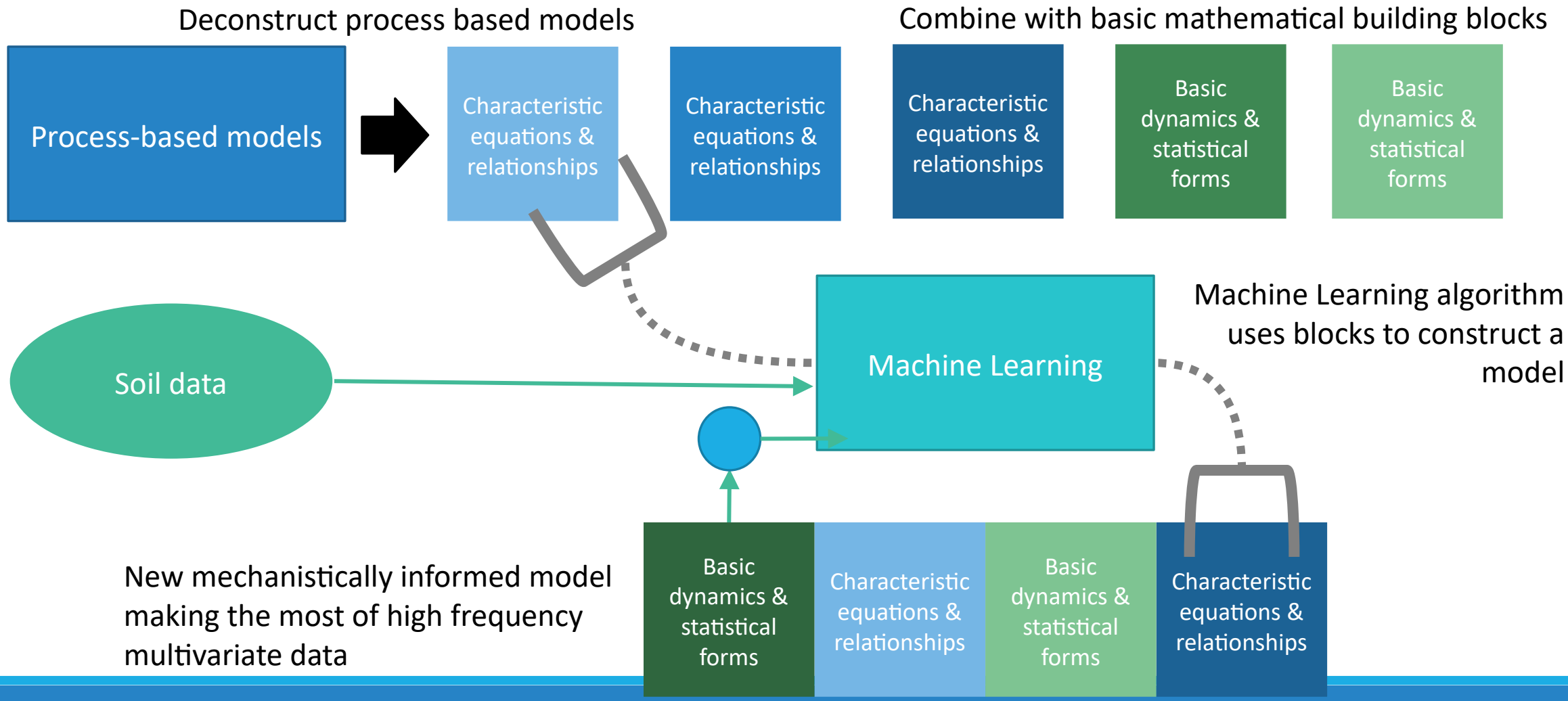
Process-based models

- Simulate change over time
- Incorporate our mechanistic understanding
- Allow us to predict under new conditions
- Highly parameterized and uncertain
- Slow and less flexible
- Not making the best of multivariate high frequency data

Machine Learning

- Help us make the most of multivariate data
- Only as good as the training dataset: needs high volumes of data under many conditions
- Current high volume soil data is short-term: limits our capacity to predict soil change

Mechanistically informed machine learning algorithms – best of both worlds



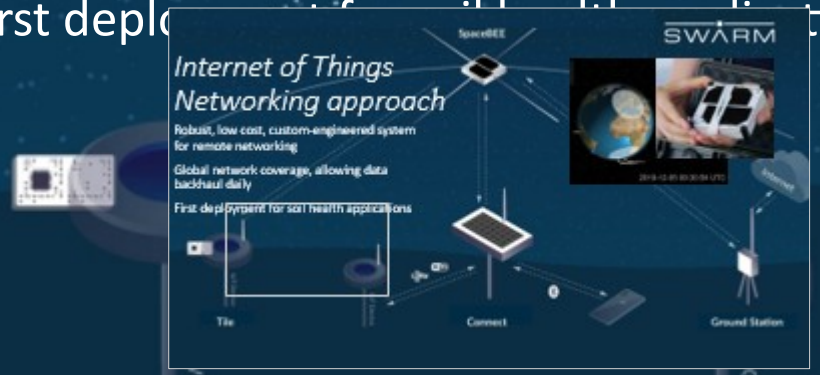
3.Scaling up

Internet of Things Networking approach

Robust, low cost, custom-engineered system for remote networking

Global network coverage, allowing data backhaul daily

First deplo[illegible]tions



SpaceBEE

SWARM



Internet

LoRa

Wi Fi

IoT Device

Tile

Connect

Ground Station

Data science challenges

1. Making sense of multi-variate data – often with varied quality

2. Dealing with heterogeneity, long- and short- term dynamics

3. Adaptive in-field data processing, making the most data transfer from field to cloud